



Practical-10 Pairs by frequency

```
from pyspark.sql import SparkSession
from pyspark.sql.function import collect_set, col
from itertools import combinations
```

```
spark = SparkSession.builder \
    .appName("Amazon Food Frequent Pairs") \
    .getOrCreate()
```

```
data = spark.read.csv("Amazon_Food_Reviews.csv",
    header=True, inferSchema=True)
```

```
reviews = data.select("UserId", "ProductId").drop
```

```
user_products = reviews.groupBy("UserId") \
    .agg(collect_set("ProductId").alias(
        "products"))
```

```
def generate_pairs(products:
    return list(combinations(sorted(products),
        2))
```

```
pair_rdd = user_products.rdd.flatMap(lambda
    row: generate_pairs(row['products']))
```

```
pairs_df = pair_rdd.map(lambda pair: pair[1]) \
    .toDF(["pair", "count"])
```



```
pair_counts = pairs_df.groupby("pair").sum(  
    count()) \  
    .withColumnRenamed("sum(count)",  
        "frequency")
```

```
frequent_pairs = pair_counts.filter(col("frequency"  
    > 1) \  
    .orderBy(col("frequency").desc)
```

```
frequent_pairs.write.csv("output / frequent_  
    product_pairs", header=True,  
    mode="overwrite")
```

```
spark.stop()
```